

Computer Assignment 2

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1 Exercise 2:1

1.1 Exercise 2:1.1

The first exercise fits a logistic regression model where getting periodontitis is the response variable (p_x), and the regular use of dental floss is the predictor variable (x). The model is defined as:

$$\text{logit}(p_x) = \beta_0 + \beta_1 x$$

In order to build up such a logistic regression model, we first input data and create a data frame.

	use	per	n
1	no	no	265
2	no	yes	148
3	yes	no	75
4	yes	yes	22

Table 1: Data Frame of Sample.

Here, *use* denotes whether an individual uses dental floss, and *per* denotes whether the individual is diagnosed with periodontitis.

Substituting the sample value into the logistic regression model above, we obtain the estimated parameters

$$\hat{\beta}_0 = -0.5825, \quad \hat{\beta}_1 = -0.6439.$$

Thus, the estimated logistic regression model is

$$\text{logit}(p_x) = -0.5825 - 0.6439 x.$$

When $x = 0$, we have $\beta_0 = \text{logit}(p_0) = \log\left(\frac{p_0}{1-p_0}\right)$;

And when $x = 1$, $\beta_1 = \text{logit}(p_1) - \text{logit}(p_0) = \log\left(\frac{p_1}{1-p_1}\right) - \log\left(\frac{p_0}{1-p_0}\right) = \log(\text{odds ratio})$.

Here, p_0 refers to $P = (\text{periodontitis}|x = 0)$; p_1 refers to $P = (\text{periodontitis}|x = 1)$.

Moreover, slightly transform the model-defining formula, we can calculate the estimated probability value:

$$p_x = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Some key values of this logistic regression model are as below:

Sign	β_0	β_1	$e^{\beta_0} = \text{Odds0}$	$e^{\beta_0+\beta_1} = \text{Odds1}$	$e^{\beta_1} = \text{OR}$	p_0	p_1
Value	-0.5825	-0.6439	0.5585	0.2933	0.5252	0.3584	0.2268

Table 2: Signs and values of logistic model1.

Answer to the questions:

1. Both β_0 and β_1 can be estimated. And we got the estimated value that β_0 is -0.5825 and β_1 is -0.6439.
2. β_0 can be interpreted as the estimated **log-odds** of developing periodontitis among those who do not regularly use dental floss. And here in this case-control study, it reflects the log-odds within the sampled population, and not necessarily the true log-odds in the general population; β_1 can be interpreted as the **log-odds ratio** of developing periodontitis for dental floss users relative to non-users.
3. Thus, we come to the conclusion that, using dental floss regularly will lead to a lower risk of getting periodontitis. Specifically, people who regularly use dental floss have approximately 0.5252 times the odds of developing periodontitis compared with those who do not.

1.2 Exercise 2:1.2

The second sub-question reverses the roles of the variables to investigate the probability of using dental floss among individuals with and without periodontitis. The model is defined as:

$$\text{logit}(p_y) = \gamma_0 + \gamma_1 y$$

Substituting the sample value into the second logistic regression model, we obtain the estimated parameters

$$\hat{\beta}_0 = -1.2622, \quad \hat{\beta}_1 = -0.6439.$$

Thus, the estimated logistic regression model is

$$\text{logit}(p_y) = -1.2622 - 0.6439 y.$$

When $y = 0$, we have $\gamma_0 = \text{logit}(p_0) = \log\left(\frac{p_0}{1-p_0}\right)$;

And when $y = 1$, $\gamma_1 = \text{logit}(p_1) - \text{logit}(p_0) = \log\left(\frac{p_1}{1-p_1}\right) - \log\left(\frac{p_0}{1-p_0}\right) = \log(\text{odds ratio})$.

Here, p_0 refers to $P = (\text{using dental floss} | y = 0)$; p_1 refers to $P = (\text{using dental floss} | y = 1)$.

The probability of using dental floss among two types of people can be calculated by

$$p_y = \frac{1}{1 + e^{-(\gamma_0 + \gamma_1 y)}}$$

Some key values of the second logistic regression model are as below:

Sign	γ_0	γ_1	$e^{\gamma_0} = \text{Odds0}$	$e^{\gamma_0+\gamma_1} = \text{Odds1}$	$e^{\gamma_1} = \text{OR}$	p_0	p_1
Value	-1.2622	-0.6439	0.2830	0.1487	0.5252	0.1294	0.2206

Table 3: Signs and values of logistic model2.

Answer to the questions:

1. Both γ_0 and γ_1 can be estimated. We got the estimated value that γ_0 is -1.2622 and γ_1 is -0.6439.

2. γ_0 can be interpreted as the log-odds of using floss among people without periodontitis; γ_1 can be interpreted as the log-odds ratio of using dental floss regularly for people with periodontitis relative to those without.
3. Thus, We can come to the conclusion that, people without periodontitis show more tendency of using dental floss regularly. Specifically, people with periodontitis have approximately 0.5252 times the odds of using floss regularly compared with those who without.

1.3 Exercise 2:1.3

From discussion above, we know that β_1 and γ_1 represent the log-odds ratio of each model respectively, so in this part we use actual data to express $\log(OR)$.

The table version of actual numbers is

use per	yes	no	sum
yes	22	75	97
no	148	265	413
sum	170	340	510

Table 4: Actual data table

For model1:

$$p_0 = P(Y = 1|X = 0), 1 - p_0 = P(Y = 0|X = 0);$$

$$p_1 = P(Y = 1|X = 1), 1 - p_1 = P(Y = 0|X = 1).$$

Thus,

$$\beta_0 = \ln \left(\frac{p_0}{1 - p_0} \right) = \ln \left(\frac{n_{01}/n_{0+}}{n_{00}/n_{0+}} \right) = \ln \left(\frac{n_{01}}{n_{00}} \right),$$

and

$$\beta_1 = \ln \left(\frac{p_1}{1 - p_1} \right) - \ln \left(\frac{p_0}{1 - p_0} \right) = \ln \left(\frac{n_{11}}{n_{10}} \right) - \ln \left(\frac{n_{01}}{n_{00}} \right) = \ln \left(\frac{n_{11}/n_{10}}{n_{01}/n_{00}} \right).$$

Corresponding to the actual value, $n_{00} = 265, n_{01} = 148, n_{10} = 75, n_{11} = 22$.

Substituting actual values,

$$\beta_0 = \ln \left(\frac{148}{265} \right) = -0.5825; \quad \beta_1 = \ln \left(\frac{22/75}{148/265} \right) = -0.6439.$$

The result **aligns with** numbers shown in Table 2.

For model2:

$$p_0 = P(X = 1|Y = 0), 1 - p_0 = P(X = 0|Y = 0);$$

$$p_1 = P(X = 1|Y = 1), 1 - p_1 = P(X = 0|Y = 1).$$

Thus,

$$\gamma_0 = \ln \left(\frac{p_0}{1 - p_0} \right) = \ln \left(\frac{n_{10}/n_{+0}}{n_{00}/n_{+0}} \right) = \ln \left(\frac{n_{10}}{n_{00}} \right),$$

and

$$\gamma_1 = \ln \left(\frac{p_1}{1 - p_1} \right) - \ln \left(\frac{p_0}{1 - p_0} \right) = \ln \left(\frac{n_{11}}{n_{01}} \right) - \ln \left(\frac{n_{10}}{n_{00}} \right) = \ln \left(\frac{n_{11}/n_{01}}{n_{10}/n_{00}} \right).$$

Corresponding to the actual value, $n_{00} = 265, n_{01} = 148, n_{10} = 75, n_{11} = 22$.
Substituting actual values,

$$\gamma_0 = \ln \left(\frac{22}{148} \right) = -1.2622; \quad \gamma_1 = \ln \left(\frac{22/148}{75/265} \right) = -0.6439.$$

The result **aligns with** numbers shown in Table 3. We can also come to the conclusion that $\beta_1 = \gamma_1$.

2 Exercise 2:2

log(dose)	-7.60	-6.22	-4.60	-3.00	-1.39	0.92	Σ
Tumor	1	2	4	9	12	32	60
No Tumor	17	17	24	23	16	8	105
Σ	18	19	28	32	28	40	105

Table 5: Given data

2.1 Exercise 2:2.1

The calculation formulas of risk and logodds are shown below:

$$\text{Risk}_i = \frac{\text{Tumor}_i}{(\text{Tumor}_i + \text{No Tumor}_i)}$$

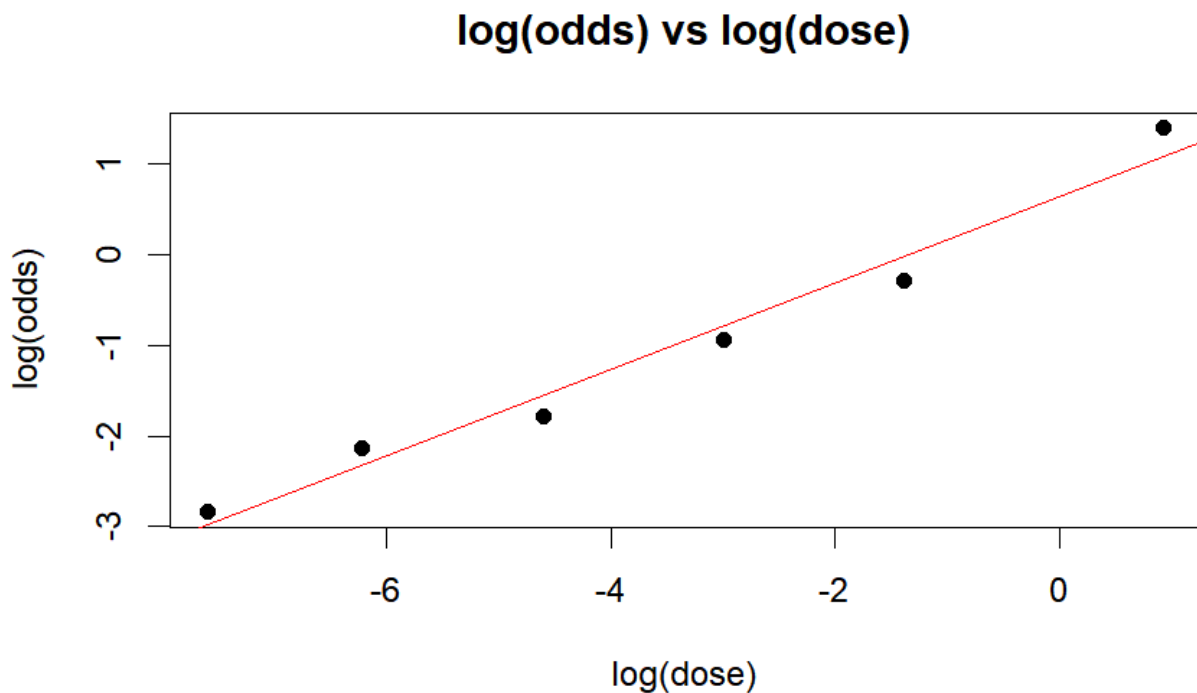
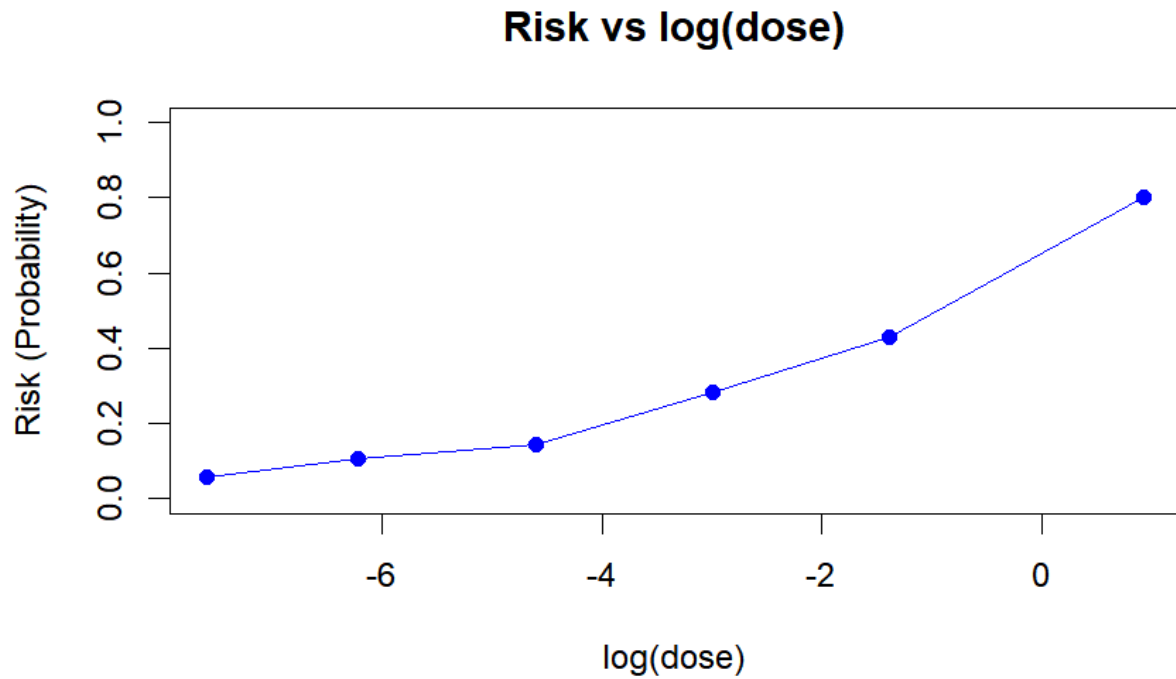
$$\log(odds) = \log\left(\frac{Risk_i}{1 - Risk_i}\right)$$

Therefore, we can calculate the risk and logodds as shown in Table 6.

log(dose)	-7.60	-6.22	-4.60	-3.00	-1.39	0.92
risk	0.056	0.105	0.143	0.281	0.429	0.800
log odds	-2.833	-2.140	-1.792	-0.938	-0.288	1.386

Table 6: Calculated results for risk and log odds

Then we make a plot of the risk against log(dose) and plot the relationship between risk and log(dose).



From the plot of Risk vs log(dose), it can be seen that the relationship between logarithmic dose and risk is nonlinear, so it is inappropriate to fit a linear model directly. However, after calculating the log-odds, the plot of log(odds) against log(dose) reveals an approximately linear relationship between the log-odds and the logarithmic dose. This linearity is a core assumption of the logistic regression model. Therefore, the Logistic Regression is suitable for this data because the log-odds and log(dose) show a linear relationship.

2.2 Exercise 2:2.2

Fit the logistic regression model and interpret the parameter estimates.

$$\text{Logistic regression : } P(Y = 1|X = x) = \pi(x) = \frac{\exp(\beta_0 + \beta_1 x)}{1 + \exp(\beta_0 + \beta_1 x)}$$

By establishing a logistic regression model, we obtain the parameter estimates shown in Table 7.

	Estimate	Std. Error	z value	Pr
Intercept	0.68702	0.25755	2.668	0.00764
logdos	0.52037	0.08502	6.120	9.35e-10

Table 7: The results of logistic regression model

Here, the intercept(β_0) is 0.68702 and the slope(β_1) is 0.52037, representing the baseline of log odds and the effect of log dose, respectively. The detailed explanation and proof shown below.

1. Intercept($\beta_0 = 0.68702$): When $\log(\text{dose}) = 0$, the log-odds of tumor occurrence is 0.68702, corresponding to a probability of $P(Y = 1|X = 0) = \pi(0) = \frac{\exp(\beta_0)}{1 + \exp(\beta_0)} = \frac{\exp(0.68702)}{1 + \exp(0.68702)} = 0.6653$.

2. Slope parameter($\beta_1 = 0.52037$): For each one-unit increase in $\log(\text{dose})$, the log-odds of tumor occurrence increases by 0.52037, which corresponds to an increase in odds by 68.3%.

$$\begin{aligned} \log OR &= \log\left[\frac{\pi(x+1)}{1 - \pi(x+1)} / \frac{\pi(x)}{1 - \pi(x)}\right] \\ &= \text{logit } \pi(x+1) - \text{logit } \pi(x) \\ &= \beta_0 + \beta_1(x+1) - (\beta_0 + \beta_1 x) \\ &= \beta_1 \\ &= 0.52037 \end{aligned}$$

$$OR = \exp(\log OR) = \exp(0.52037) = 1.6826$$

3. Statistical significance: The slope of the model is significant ($p < 0.001$), indicating a significant positive effect of dose on tumor risk, with the risk increasing as the dose increases.

2.3 Exercise 2:2.3

Find covariance matrix, parameter correlation, confidence intervals, and risk at specific dose.

Using `vcov(model22)`, we obtain the covariance matrix for the estimates as shown in Table 8.

	Intercept	logdose
Intercept	0.06633368	0.014358467
logdose	0.01435847	0.007229089

Table 8: Covariance matrix for the estimates

Then we could calculate the correlation coefficient: $r = \frac{\text{Cov}(\text{intercept}, \text{logodds})}{\sigma_{\text{intercept}} \sigma_{\text{logodds}}} = \frac{0.014358467}{\sqrt{0.06633368 \times 0.007229089}} \approx 0.655691$, indicating that the correlation between intercept and slope is positively related and relatively strong.

	Lower boundary (2.5%)	Upper boundary (97.5%)
Intercept	0.1822292	1.1918194
logdos	0.3537212	0.6870093

Table 9: Confidence intervals for the parameters

Table 9 represents the confidence intervals for the parameters, where 95% confidence intervals for the intercept is [0.182, 1.192], and 95% confidence intervals for logdose is [0.354, 0.687]. And when the log(dose)=-1.39, the 95% confidence interval for the risk at that level is [0.394, 0.588]. The calculation formulae for the risk at that level are shown as follows.

$$x = (1, -1.39)^T; \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}; \Sigma = \begin{pmatrix} Var(\beta_0) & Cov(\beta_0, \beta_1) \\ Cov(\beta_0, \beta_1) & Var(\beta_1) \end{pmatrix}.$$

Here, β is the estimate of coefficients and Σ is the same with r output function(vcov).

$$\text{logit } \pi(x) = x^T \beta; SE(\text{logit } \pi(x)) = \sqrt{x^T \Sigma x}.$$

So for logit we have 95% confidence interval for it:

$$[\text{logit } \pi(x) - 1.96SE(\text{logit } \pi(x)), \text{logit } \pi(x) + 1.96SE(\text{logit } \pi(x))] = [p_{low}, p_{high}]$$

Then we can get the 95% confidence interval for tumor risk at dose 0.25 (log(dose)=-1.39) is

$$\left[\frac{\exp(p_{low})}{1 + \exp(p_{low})}, \frac{\exp(p_{high})}{1 + \exp(p_{high})} \right] = [0.3940875, 0.5884584]$$

2.4 Exercise 2:2.4

Perform a Wald-test

$$H_0 : \beta_1 = 0; H_a : \beta_1 \neq 0.$$

From the results of the logistic regression model built in Exercise 2:2.2, we obtain the slope estimate(β_1) is 0.52037 and the standard error of β_1 is 0.08502. Then we can construct the Wald statistic:

$$z_W = \hat{\beta}_1 / \sqrt{\text{Var}(\hat{\beta}_1)} \sim \mathbf{N}(0, 1)$$

The calculated Wald statistic is $Z_W = 6.1202$, and we get the p-value: $p = 2 \times (1 - \Phi(|Z_W|)) = 2 \times (1 - \Phi(6.1202)) = 9.3451e - 10 < 0.001$. That means by using Wald test we reject the null hypothesis $\beta_1 = 0$, i.e. the log(dose) do have effect on tumor risk.

2.5 Exercise 2:2.5

Investigate the relationship between sample size and variance of parameter estimates

Original value									
log(dose)	-7.6	-6.22	-4.6	-3	-1.39	0.92		Intercept	logdose
x	1	2	4	9	12	32	Intercept	0.06633368	0.01435847
n	18	19	28	32	28	40	logdose	0.01435847	0.00722909
Mutiply 10									
log(dose)	-7.6	-6.22	-4.6	-3	-1.39	0.92		Intercept	logdose
x	10	20	40	90	120	320	Intercept	0.00663337	0.00143585
n	180	190	280	320	280	400	logdose	0.00143585	0.00072291
Mutiply 100									
log(dose)	-7.6	-6.22	-4.6	-3	-1.39	0.92		Intercept	logdose
x	100	200	400	900	1200	3200	Intercept	0.00066334	0.00014358
n	1800	1900	2800	3200	2800	4000	logdose	0.00014358	0.00007229
Mutiply 1,000									
log(dose)	-7.6	-6.22	-4.6	-3	-1.39	0.92		Intercept	logdose
x	1000	2000	4000	9000	12000	32000	Intercept	0.00006633	0.00001436
n	18000	19000	28000	32000	28000	40000	logdose	0.00001436	0.00000723

Table 10: Table counts and Covariance Matrix

Table 10 shows original values and the counts of table and covariance matrix for each time multiply 10 repeatedly. The left part shows the count of each time and the right part shows the covariance matrix. The variance of a parameter estimate is generally inversely proportional to the sample size. This is also true here. By multiplying all counts in table by 10 repeatedly, we can see that the variance for each parameter are getting smaller and the variance is also ten times smaller than the previous one.